Centre for Mathematical Sciences, CVML, September 2026

Image Analysis, Assignment 2

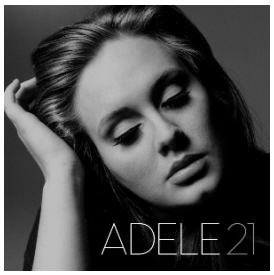
Emphasis in the report. The purpose of the assignment is not only to obtain the requested numerical results, plots, and implementations, but also to demonstrate that you understand why the methods behave as they do. Keep the report concise: use calculations and code to support your conclusions, and place particular emphasis on predictions, interpretation of results, comparison of methods, assumptions, and possible failure modes.

1 Filtering

Shown below is an image and the results of convolving this image with five different (convolution) filters. Combine each convolved image with the correct filter.

$$f_1 = \frac{1}{6}\begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \qquad f_2 = \frac{1}{12}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad f_3 = \frac{1}{3}\begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

$$f_4 = \frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{bmatrix} \qquad f_5 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad f_6 = \frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$



Original



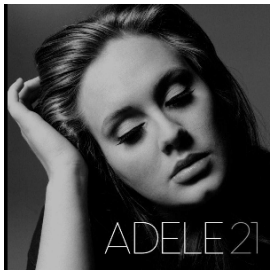
A



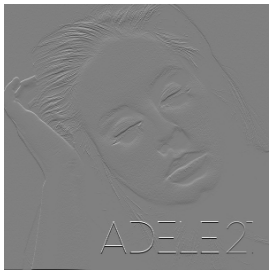
B



C



D



E

For the report:

- Match each output image to the corresponding filter and motivate each match from the structure of the filter, not only from visual similarity.
- Before making the final matching, classify the filters qualitatively (for example, smoothing, sharpening, edge/difference detection, or spatial shift) and state what visual effect you expect from each one.
- For at least two of the filters, explain how the signs, symmetry, and sum of the filter coefficients support your interpretation.
- Identify one match that is especially easy to distinguish and one that is potentially more ambiguous, and explain why.

2 Interpolation

Assume a one-dimensional image (or signal)

$$f = \begin{bmatrix} 1 & 2 & 2 & 4 & 7 & 6 & 5 \end{bmatrix}, \quad \text{(at points } i = 1, \ldots, 7\text{)}.$$

a)

- Explain what linear interpolation means.
- Sketch the linear interpolation $F_{lin}(x)$ of f for $1 \leq x \leq 7$.
- Is $F_{lin}(x)$ continuous? Is $F_{lin}(x)$ differentiable?

b) The interpolation can be expressed using the following equation

$$F_g(x) = \sum_{i=1}^{7} g(x-i) f(i), \tag{1}$$

where the function $g(x)$ is different for different types of interpolation.

- Which function $g(x)$ corresponds to linear interpolation?
  Give the function $g(x)$ for $-4 \leq x \leq 4$.

c) Let $g(x)$ be defined as

$$g(x) = \begin{cases} \frac{6}{7}(1 - |x|^2) & \text{if } |x| \leq 1, \\ -\frac{6}{7}(|x|^3 - 5|x|^2 + 8|x| - 4) & \text{if } 1 < |x| \leq 2, \\ 0 & \text{if } |x| > 2. \end{cases}$$

- Plot $g(x), -3 \leq x \leq 3$.
- Plot the interpolation $F_g(x)$ of the function f using this $g(x)$, for $1 \leq x \leq 7$ (e.g. using Python or Matlab for the calculations).
- Is $F_g(x)$ continuous?
- Is $F_g(x)$ differentiable? (Here you need to have an argument that is better than "you can see it in the graph". Hint: the properties of $F_g(x)$ will depend on the properties of $g(x)$. If a function is piece-wise defined by functions that are differentiable within the intervals, what do you need to check to ensure that the whole function is differentiable?).

For the report:

- Provide the requested sketches, plots, expressions, and conclusions, but use the derivations to explain the behavior of the interpolants rather than only reporting formulas.
- For linear interpolation, explain where continuity and differentiability come from and identify where differentiability can fail.
- For the given piecewise function g, inspect the pieces before plotting and state what continuity and differentiability properties you expect at the joining points. Verify these expectations analytically.
- Compare the interpolation obtained with this g with the linear interpolation: comment on locality, smoothness, and how values between the samples are influenced by neighboring samples.
- Use at least one specific interval or sample location from the given signal to connect the analytical properties of g to a visible feature in your plot of F_g.

3 Classification using Nearest Neighbour and Bayes theorem

As output from an imaging system we get a measurement that depends on what we are seeing. For three different classes of objects we get the following measurements.
Class 1:

0.4003, 0.3985, 0.3998, 0.3997, 0.4015, 0.3995, 0.3991

Class 2:

0.2554, 0.3139, 0.2627, 0.3802, 0.3247, 0.3360, 0.2974

Class 3:

0.5632, 0.7687, 0.0524, 0.7586, 0.4443, 0.5505, 0.6469

3.1 Nearest Neighbours

Use nearest neighbour classification. Assume that the first four measurements in each class are used for training and the last three for testing. How many measurements will be correctly classified ?

3.2 Gaussian distributions

Assume that the measurements have normal distribution with known parameters (expected value m_i and variance σ^2). For class 1: $m_1 = 0.4$, $\sigma_1 = 0.01$. For class 2: $m_2 = 0.32$, $\sigma_2 = 0.05$. For class 3: $m_3 = 0.55$, $\sigma_3 = 0.2$. All classes are as likely to occur. What is the maximum a posteriori classification of a measured x, if all measurements are independent? How many measurements are correctly classified? In this case you can try

to classify all seven samples in each class, since no measurements were used for training. Hint: In Python there's a function in Scipy called `scipy.stats.norm.pdf` and Matlab has a function called `normpdf`.

For the report:

- Give the classification results for both methods and show the detailed calculation for at least one example for nearest-neighbour classification and one example for Gaussian/MAP classification.
- Before calculating all classifications, inspect the three groups of measurements and their Gaussian parameters. State which class you expect to be easiest and which you expect to be hardest to classify, and explain why.
- Compare the errors made by nearest neighbour and by the Gaussian model. For any misclassified sample, explain what in the training data or class distributions makes that error plausible.
- Explain conceptually why the two classifiers can give different decisions for the same measured value even though both are using the same class labels.
- Briefly discuss how the MAP rule would change if the classes were not equally likely, without changing the numerical priors in this task.

4 Image Classification

A scene is generated by some random mechanism. By studying a large number of noise free realizations of the scene we have that (for the three possible cases):

$$\begin{matrix}0\ 1\\1\ 0\end{matrix} \text{ has prob } \frac{1}{4}; \qquad \begin{matrix}1\ 0\\0\ 1\end{matrix} \text{ has prob } \frac{1}{2}; \qquad \begin{matrix}1\ 1\\1\ 0\end{matrix} \text{ has prob } \frac{1}{4}$$

(a priori probabilities). One of these images has been distorted by noise in the sense that the value of a pixel has been changed with the probability ϵ, i.e. Prob (observing 0 | the correct value is 1) = P (observing 1 | the correct value is 0) = ϵ. Assume that noise in different pixels is independent. Now consider the image

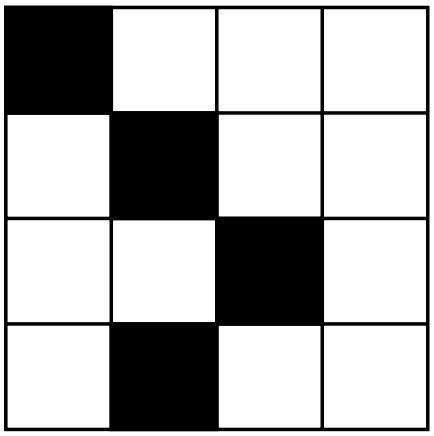
$$\begin{matrix}0\ 1\\1\ 1\end{matrix}$$

For the report:

- Using Bayes theorem, calculate the posterior probabilities and the MAP estimate of the scene for (a) $\epsilon = 10\%$ and (b) $\epsilon = 50\%$. Show the answers in percentages or percentage decimals.
- Before normalizing the posteriors, explain how each observed pixel contributes to the likelihood of a candidate scene under the independent-noise assumption.
- Compare the two noise levels. Explain why changing ϵ changes the balance between the observed image and the a priori probabilities.
- Comment on what the result for $\epsilon = 50\%$ says about the amount of information carried by the observed pixel values in this noise model.

5 Line Classification

Consider a binary 4×4 image of a scene with a vertical line. In the *correct* image all pixels would be white except one vertical row with black pixels. Unfortunately the camera used is far from perfect. Errors in different pixels are independent with p(white | line)=p(black | not line)=ϵ and consequently, Prob(black | line)=Prob(white | no line)=$1 - \epsilon$. Assume the a priori probability for the line to be located in column 1 or 4 is 0.3 (each) and the the a priori probability that the line is in column 2 or 3 is 0.2 (each). Calculate the maximum a posteriori estimation of the following image when $\epsilon = 0.2$.



For the report:

- Give the a priori probabilities of the four possible line images, calculate the corresponding posterior probabilities, and state the MAP estimate. Show the answers in percentages or percentage decimals.
- Before carrying out the full calculation, inspect the observed 4×4 image and predict which line locations appear most plausible from the pixel evidence alone. Then compare this with the final posterior ranking.
- For at least two candidate line positions, explain which observed pixels provide evidence for and against the candidate.
- Explain how the unequal priors for the outer and inner columns influence the final decision. Would the pixel evidence and the prior information favour the same candidate in this example?

6 Character Classification

An image (with 5×3 pixels) has probability 0.3 of being a 'B' (image ω_1), 0.45 of being a '0' (image ω_2) and 0.25 of being an '8' (image ω_3). In the final image there is also measurement noise. The probability of error is bigger for white pixels. An truly white pixel has a probability 0.35 of being wrongly measured as black in x. An truly black pixel has a probability 0.2 of being wrongly measured as white in x. How should the image x below be classified using maximum a posteriori probabilities (MAP) in the classification criterion.

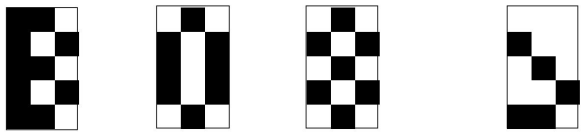


Figure 1: Mean images for three different classes, and an observed image x.

For the report:

- Give the a priori probabilities of the three character images, calculate the posterior probabilities, and state the MAP classification. Show the answers in percentages or percentage decimals.
- Explain how the likelihood of the observed image x is constructed from the individual pixels and why mismatching a white pixel and mismatching a black pixel do not have the same effect in this model.
- Before normalizing the posteriors, identify which parts of the observed pattern give the strongest evidence for or against each of the three classes.
- Compare the influence of the prior probabilities with the influence of the pixel evidence. Is the final classification mainly determined by the prior or by the observed pattern? Motivate your answer using the calculated quantities.

7 The Garment Classification system - part 2 - Feature extraction

During all of the four assignments you are gradually going to build and test a small system for classification of different clothing items. During this second assignment your task is to write a function `segment2features` that takes a segmented image and returns a feature vector, and to test this function on a few images using a benchmark script.

On the Canvas page for the course there is a zip-file with data. By downloading and unpacking the file you will obtain a folder `garment_recognition_project` and in this folder there is a folder
`datasets`
In the datasets folder there is for now only one folder 'short1', which contains a few test examples and ground-truth both for segmentation and for recognition. Later on we will add more folders with additional (and more challenging images).

Given a binary image B (thus only with zeros and ones), study the region of pixels that are equal to one. Use your imagination to define *at least, possibly more than* six different features (numbers), that can be used to classify which garment class the region corresponds to. It is good if the feature (the value) does not depend on which position the region has. In other words, if the regions is shifted in the x or y direction, then the same features will be obtained.

Suggestions: One could try using moments, histograms, the sum of the pixel values along a row or in a subregion, etc. You can also think about the filter bank idea from the lecture on features. Remember that it's difficult to know which features that work well, without testing. To remove the influence of the position one could first translate the image so that the centre of mass is in the origin. Normalizing features so that they all have approximately the same range of values is also often a good idea (e.g. use values between 0 and 1 or -1 and 1).

Specific instructions

In the project folder there is a script `assignment2.py`. Study the script that reads an example image and runs the function `im2segment` on it, and then the function `segment2feature`. The function `segment2feature` takes an image from the segmentation, and returns a vector of features. Replace these functions with your own implementations. The script also runs the benchmark function `benchmark_assignment2` that tests your function on the dataset.

For the report

For the report:

- Describe and motivate the features you have chosen. For each feature (or group of closely related features), state what property of a garment it is intended to capture and what invariances, such as translation invariance, you expect it to have.
- Before running the benchmark, formulate at least two concrete expectations about the feature vectors: one concerning different images of the same garment class and one concerning images from different classes.
- Supply your code and representative examples of input segments with their corresponding feature vectors, including both different images of the same class and examples from different classes.
- Use these examples and the benchmark visualization/results to assess your expectations. Identify features that appear discriminative and features that vary more than you expected within a class.
- Discuss at least one plausible failure mode: give an example of two garment shapes that could obtain similar feature vectors, or one garment whose feature vector could change substantially because of shape, scale, rotation, segmentation error, or another relevant variation.
- If you were allowed to add or replace one feature after inspecting the results, explain what you would change and what weakness in the current representation you would try to address.